
Beyond Raw Expression: Geneformer and GenePT Embeddings as Features for Immune Checkpoint Inhibitor Response Prediction

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Abstract

Predicting patient response to immune checkpoint inhibitors (ICIs) from single-cell RNA sequencing (scRNA-seq) data remains a significant challenge due to technical variance across sequencing platforms and limited sample sizes. A key ICI response prediction pipeline, PRECISE, has demonstrated success in using raw gene expression values and manual feature selection. However, its performance on new cancer types was not as successful. In this study, we propose and evaluate GG-Boost (Geneformer/GenePT-XGBoost), a pipeline that replaces raw expression features with pretrained embeddings from single-cell foundation models. We compare two foundation models: Geneformer, which utilizes rank-based genomic encodings, and GenePT, which leverages semantic, literature-based gene representations. Our results demonstrate that both models achieve high validation performance on melanoma training data (AUC 0.901 and 0.905). However, in cross-cancer transfer to a basal cell carcinoma (BCC) dataset, performance diverged significantly. Geneformer-Boost failed to generalize (AUC 0.33), likely due to sensitivity to sequencing depth differences between the sequencing platforms used on the two cancer datasets. GenePT-Boost achieved an AUC of 0.80, outperforming the PRECISE 11-gene signature baseline (AUC 0.68). Despite the high accuracy, we observed a narrow probabilistic spread for individual cell predictions, centered around 0.5, likely due to non-informative cells preventing the model from learning meaningful decision boundaries. Our findings suggest that there is potential for foundation models to improve the generalizability of a model trained on one cancer type to others, but single cell data requires processing to ensure immunotherapy response signals are not diluted. This approach highlights the power of foundation models in enhancing the prediction of treatment response across cancer types.

1 Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled characterization of gene expression and analysis of heterogeneity within tumour tissue samples at the cellular level [1]. This technology is particularly advantageous in the study of cancer immunotherapy, where the individual cells of the tumour microenvironment (TME) can be analyzed to uncover patterns of gene expression that could possibly infer patient resistance or susceptibility to different treatments [2]. Notably, immune checkpoint inhibitors (ICIs), a popular immunotherapy treatment in melanoma patients, produces a durable response in only a fraction of patients [3]. With high throughput sequencing technologies becoming more abundant and more accurate, there is an increasing need to analyze the single-cell data in such a way that candidate patients for immunotherapy can be selected without the need for gruesome analysis of their entire genome or subjecting them to multiple different treatments.

ICIs work by binding to receptors involved in T cell attenuation, which induces an enhanced anti-tumour T cell response [4, 5]. Such therapies can either be anti-CTLA4 or anti-PD1, specifying the name of the receptor they target [4]. These therapies primarily act in the TME, which allows for any expressed receptors and proteins to be relevant predictive biomarkers in treatment response [5].

One of the biggest challenges in the application of machine learning (ML) to ICI response prediction is the relatively limited number of samples from which the response label of the patient is derived, but the large number of single-cells available to train on [6]. This creates the need for a multi-step prediction approach, where response labels are propagated from the sample level to individual cells, then predictions are made at the cellular level and results are aggregated back to a single patient-level score which reflects the predicted response to ICI therapy.

We built our pipeline off the PRECISE framework developed by Pinhasi & Yizhak which predicts immune response from scRNA-seq data using XGBoost [6]. A key strength of their approach is its interpretability; by using SHAP values, they identified not only which genes were most predictive, but also the direction of their contribution. This revealed that many highly predictive features aligned with genes already known to be biologically relevant in immune pathways and play a role in cancer development, lending confidence to the model’s predictive abilities.

However, the model falls short in its use of raw expression data which can differ across cancer types and sequencing platforms, making it infeasible to run the classifier on other cancer data sets with different features. To mitigate this limitation, the authors distilled their predictions to an 11-gene signature, a compact set of genes whose expression alone was sufficient to predict ICI response. This signature was applied across multiple cancer types, including triple-negative breast cancer, non-small cell lung cancer, and basal cell carcinoma (BCC), demonstrating that the predictive signal captured in melanoma T cells reflects broader pan-cancer immune response mechanisms. PRECISE succeeded in generalizing across some cancer types, but also saw a reduction in accuracy on others, leaving room for improvement.

Another limitation is that their model outputs binary scores for each cell, so the sample level score reflects the proportion of responder cells, but does not factor in the model’s confidence in its prediction (a cell labelled responder with 0.51% confidence weighted the same as a cell labelled with 99% confidence). Predictions would be easier to trust and interpret given a measure of prediction confidence. This would also allow for a biologically relevant aggregation of the cell-level predictions to a sample-level predictive label.

Our paper explores whether the use of foundation model embeddings improve PRECISE’s ability to generalize across cancer types. We propose using a foundation model to replace raw gene expression features with pretrained embeddings as input to the XGBoost classifier used in the PRECISE pipeline. These foundation models are pretrained on tens of millions of single cells spanning diverse human tissues and disease states. Exposure to a large variety of cells allows the model to learn fundamental cell features to create a general-purpose biological representation of cell state [7]. Critically, all cells, regardless of cancer type or sequencing platform used, are mapped into the same embedding space, making cross-dataset comparison more meaningful than direct comparison of raw expression vectors which are sensitive to batch effects and platform differences. Thus, using embeddings in place of raw gene expression values will allow for the use of the full classifier on new cancer datasets by providing a shared feature space that the model can be both trained and tested with new data on. We theorize that the use of the XGBoost classifier, and all the genes in the data rather than just the 11 gene signature formula will improve model performance on new cancer data sets as the full feature set may provide additional immune signals the 11 gene signature misses. The contribution of our method is in improving PRECISE’s predictive performance across cancer types through the use of foundation model embeddings.

Few studies have used foundation models in this way. One such study introduces a foundation model called COMPASS that predicts immunotherapy response using bulk tumour transcriptomes from 33 cancer types with a concept-bottleneck transformer [8]. The model generalizes well to unseen cancer types, outperforming 22 baseline methods. Moreover, COMPASS is an interpretable model that connects “gene expression to immune concepts, revealing mechanisms of response or resistance” [8]. In another approach to the use of embeddings, Liu et. al [9] developed the EGSP model, which uses bulk RNA-seq data across 25 cancer types in survival outcome prediction. Their approach built upon scFoundation [10] by appending a survival prediction head and using all three of clinical, gene expression, and embedding data at once. They found that the use of multi modal data outperformed

single-modality methods, demonstrating that embeddings provide non-redundant prognostic signals when used with raw gene expression data. Similar to COMPASS, another strength of EGSP is in its interpretability; a SHAP analysis was performed to link embeddings to biological and clinical processes.

A key limitation of both COMPASS and EGSP is in the reliance on bulk RNA-seq data. This captures average gene expression across cells, but single cell resolution is lost and key signals from rare immune cell populations are masked. Our pipeline aims to use single cell RNA-seq data to create embeddings so that the high-resolution signals of immune response are preserved rather than diluted by averaging expression levels. The pan-cancer and multimodal datasets used with COMPASS and EGSP are another limitation as they require large cancer and clinical data repositories that may not be available in real-world hospital settings. In contrast, our approach explores the use of a single cancer dataset in predicting on other cancers, reducing the amount of data required and the need to integrate various datasets, making it a more accessible framework than COMPASS or EGSP. Lastly, with the use of XGBoost, our model is also simpler than the transformer based architectures of other state of the art models, further demonstrating its ease of use in clinical settings compared to complex models.

Our pipeline, GG-Boost (Geneformer/GenePT-XGBoost), differs from other foundation model-based pipelines in the use of raw foundation model embeddings in a separate classifier – XGBoost. We use two pretrained single cell foundation models, Geneformer [11] and GenePT [12], to generate embeddings for each cell, which are then separately fed into XGBoost for predictions. Our experiments below test whether foundation model embeddings in general improve XGBoost’s predictions of immunotherapy response on new data sets and compare the performance of embeddings from GenePT and Geneformer. We evaluate GG-boost using a leave-one-out cross-validation (LOO-CV) approach on a melanoma cancer dataset, followed by external validation on a different cancer - basal cell carcinoma (BCC).

Our results demonstrate that GenePT-Boost improves the cross cancer generalizability of PRECISE’s classifier, achieving an AUC of 0.8 on the BCC cohort, compared to the 0.33 AUC achieved by Geneformer-Boost and the 0.68 AUC baseline of PRECISE’s 11 gene-signature. Despite this, cell-level and sample-level probability scores were close to a 0.5 decision threshold, indicating the lack of predictive confidence GG-Boost provides.

2 Method

(Fig. 1 I) below gives an overview of the GG-Boost pipeline. After preprocessing the melanoma scRNA-seq data, two separate embedding matrices are created with Geneformer and GenePT. These embeddings are separately used to train XGBoost, and similar basal cell carcinoma embeddings are then used to validate the models.

2.1 Data Retrieval and Preprocessing

We used publicly available scRNA-seq data from 48 tumor samples of melanoma patients who were treated using checkpoint inhibitors (GSE120575) [13]. The dataset contained 16,291 CD45+ cells and 55,736 genes. After excluding ribosomal protein genes (genes starting with ‘MT-’, ‘RPS’, ‘RPL’, ‘MRP’, and ‘MTRNR’) and genes expressed in less than 3% of cells, we were left with 12,404 genes. We additionally filtered and kept only T-cells (clusters 5, 6, 7, 8, 9, 10, and 11), for direct comparison with PRECISE’s results on T-cell filtered data. This resulted in a final number of 12,243 cells.

An additional dataset of BCC was used to validate the model (GSE123813) [14]. This scRNA-seq data was obtained from 11 patients and contained 9 T cell cluster types, with an initial 55,030 cells and 23,310 genes. After filtering by the same method described above, we were left with 33,106 cells and 9,541 genes. Although Pinhasi and Yizhak [6] validated PRECISE on six other cancers, we chose to focus on just BCC (basal-cell carcinoma) due to data-set size and platform limitations.

2.2 Geneformer Embeddings

Geneformer is a deep learning model pretrained on the single-cell transcript data of upwards of 104 million cells [15]. The model was pretrained with a self-supervised masked learning objective [11]. This is where the model learned to predict the identity of masked genes based on the content of

unmasked genes. The model takes the input data and turns it into a rank value encoding for each cell. Thus, each cell’s genes are ranked by their expression in that cell normalized by their expression across the entire Geneformer reference library. This makes it particularly suited for cross-dataset generalization. Moreover, Geneformer is able to prioritize genes that distinguish cell state.

To prepare data for input into Geneformer, gene names were first converted to Ensembl IDs. For genes mapped to multiple or duplicate Ensembl IDs, only the first was used. An additional column was added that contained the sum of TPM per cell (expected by Geneformer). This gave us a final data frame of shape 12,243 cells and 10,087 genes. Using Geneformer V2-104M, which produces 768-dimensional embeddings, we got a final output of shape 12,243 cells x 768 features.

For the BCC dataset, we used the same process but slightly altered the additional column to be a sum of raw counts (not TPM). Additionally, Geneformer’s batch size was changed from 100 to 10 to accommodate for the larger dataset. We inputted a data frame of 33,106 cells and 8,842 genes into Geneformer and got a final output shape of 33,106 cells x 768 Geneformer embeddings.

2.3 GenePT Embeddings

GenePT is a transformer based foundation model that leverages ChatGPT to create embeddings based on each gene’s NCBI (National Centre for Biotechnology Information) text description [12]. Single cell embeddings are created by calculating an average of the gene embeddings, weighted by their expression level in the cell. A key advantage of GenePT in its simplicity: unlike other foundation models, GenePT does not require millions of cells to train on as it uses existing biological knowledge.

The creators of GenePT released pre-computed Open AI embeddings for the genes used in their paper [12], which we used to avoid paying for Open AI API usage. The precomputed embeddings overlapped with 10,401 genes out of 12,404 in the melanoma data set. The missing genes were considered a satisfactory trade off for the fewer resources required to run GenePT. Since embeddings for each gene were already generated, the formation of single cell embeddings was a simple matter of averaging gene embeddings. Thus, we chose GenePT due to its ease of use.

After input into GenePT, our melanoma data set dimensions were 12,243 cells x 1,536 embedding features. Each cell used a weighted average of 10,401 of these 1,536 dimensional embeddings. The BCC dataset was also filtered for 9,284 (out of 9,541) genes found in the embeddings, resulting in a final BCC matrix of size 33,106 cells x 1,536 embedding features.

2.4 XGBoost and Sample Score

Extreme gradient boosting (XGBoost) is a gradient-boosted decision tree (GBDT) machine learning library [16]. The idea behind it is to iteratively train a collection of decision trees, using each iteration’s error residuals to fit the next model [17]. The final prediction is a weighted sum of all the tree predictions. The key feature that sets XGBoost apart from other GBDTs is that it builds the trees in parallel rather than sequentially. Additionally, it is able to output a metric related to the probability of a class label. XGBoost generally works well with structured data, especially when the number of features is less than the number of observations, but it can overfit on noisy data.

We used the melanoma and BCC metadata to attach sample names and response status to each cell. Cells labelled as responder cells were assigned 1 and non-responder cells were assigned 0. An XGBoost classifier was trained on the melanoma embeddings to predict the probability of a cell being a responder. All cell-level probabilities for a given sample were averaged to represent the confidence of the ‘responder’ label at the sample level.

Model performance was evaluated using leave-one-out cross-validation (LOO-CV) at the sample level. In each fold, all cells from one patient sample were held out as the test set while the model was trained on cell embeddings from all remaining samples. The model then produced a per-cell probability of response for each held-out cell, and calculated a sample score by averaging probabilities of response for all cells in the sample. This process was repeated for all 48 melanoma samples, and the area under the receiver operating characteristic curve (ROC-AUC) was computed across the resulting sample-level scores as the primary performance metric.

To address class imbalance between responder and non-responder cells, the ‘scale pos weight’ parameter was set to the ratio of non-responder to responder cells in each training fold. Hyperparameters

were tuned empirically through iterative evaluation of LOO-CV AUC (starting with parameters set by Pinhazi and Yizhak in their XGBoost model [6]). The final configuration for Geneformer and GenePT used ‘max depth=3’, ‘learning rate=0.01’, and ‘n estimators=25’. Shallow trees with few estimators were found to perform best. After extensive hyperparameter tuning and testing, we found that simpler models performed better on validation data than fine tuned models with many hyperparameters, so the focus for tuning was on tree structure and learning rate.

For external validation, a single XGBoost model was trained on all 48 melanoma samples using the tuned hyperparameters and applied to BCC T cell embeddings without retraining. ROC-AUC was computed on the 11 BCC patient samples to assess cross-cancer generalizability of the model.

3 Results

3.1 XGBoost Cell Type Prediction

Previously, Pinhazi and Yizhak were able to achieve an AUC of 0.84 on their base model, trained with the melanoma dataset. An additional AUC of 0.94 was achieved with their 11-gene signature, after filtering for T cells and excluding non-predictive cells [6]. We adopt part of the PRECISE framework, using the cell-level LOO-CV with XGBoost and sample-score aggregation, but we replace the raw expression features with foundation model embeddings as input into XGBoost.

The melanoma dataset consists of 48 samples treated with a combination of anti-PD1 and anti-CTLA4 therapy and sequenced using Smart-seq2 technology [13]. Following preprocessing (Methods), we were left with 3,916 cells labelled as ‘responder’ and 8,327 cells labelled as ‘non-responder’ for training, originating from 17 responder samples and 31 non-responder samples. Using Geneformer’s 768-dimensional embeddings, we trained XGBoost using LOO-CV to predict on 12,243 cells. After hyperparameter tuning, we achieved an AUC of 0.901 ($p = 5.40 \times 10^{-6}$) (Methods) (Supplementary Figure S1). We repeated this tuning and training with GenePT’s 1,536 dimensional embeddings achieving an AUC of 0.905 (Supplementary Figure S1). This is comparable to PRECISE’s performance and even better than their base model.

To further characterize Geneformer-Boost’s performance, the distribution of predicted cell proportions per sample is visualized as a stacked bar chart, with responders characterized by a probability of greater than or equal to 0.5 (Fig. 1 A). Though most samples show a higher proportion of correctly predicted cells, no sample contains a strong proportion of one cell type.

Predictions from Geneformer-Boost were outputted as a probability representing the confidence of the prediction being a ‘responder’. A sample-level predicted score was computed as the mean probability across all cells within a sample (Methods). By visualizing these scores, we noticed that scores concentrate around 0.5 ± 0.0025 (Fig. 1 B), consistent with the narrow range of cell prediction proportions, demonstrating that the model operates near decision boundaries. This indicates that the model produced near-uniform predictions with no meaningful discriminative capacity.

Although GenePT-Boost achieves a similar AUC score, the proportions of predicted responder/non responder cells reveals a difference in model confidence (Fig. 1 E). The proportions of each group range from ~0.4-0.6 for Geneformer-Boost, and ~0.1-0.9 for GenePT-Boost, illustrating GenePT-Boost’s ability to correctly classify more cells in a sample than Geneformer-Boost. Moreover, a visualization of the sample level probabilistic scores (Fig. 1 F) shows a slightly wider range of scores compared to Geneformer-Boost - 0.5 ± 0.04 . Although the probabilistic scores are still close to 0.5, the slightly wider range demonstrates that GenePT-Boost is slightly better at capturing subtle, yet stable biological signals of response.

3.2 Predicting on other cancer types

To assess the cross-cancer generalizability of the trained model, we validated the melanoma-trained model on a separate cancer type, BCC. PRECISE achieved an AUC of 0.68 (0.70 after filtering out non-predictive cells) by applying their 11-gene signature to predict the response status of samples on this dataset [6].

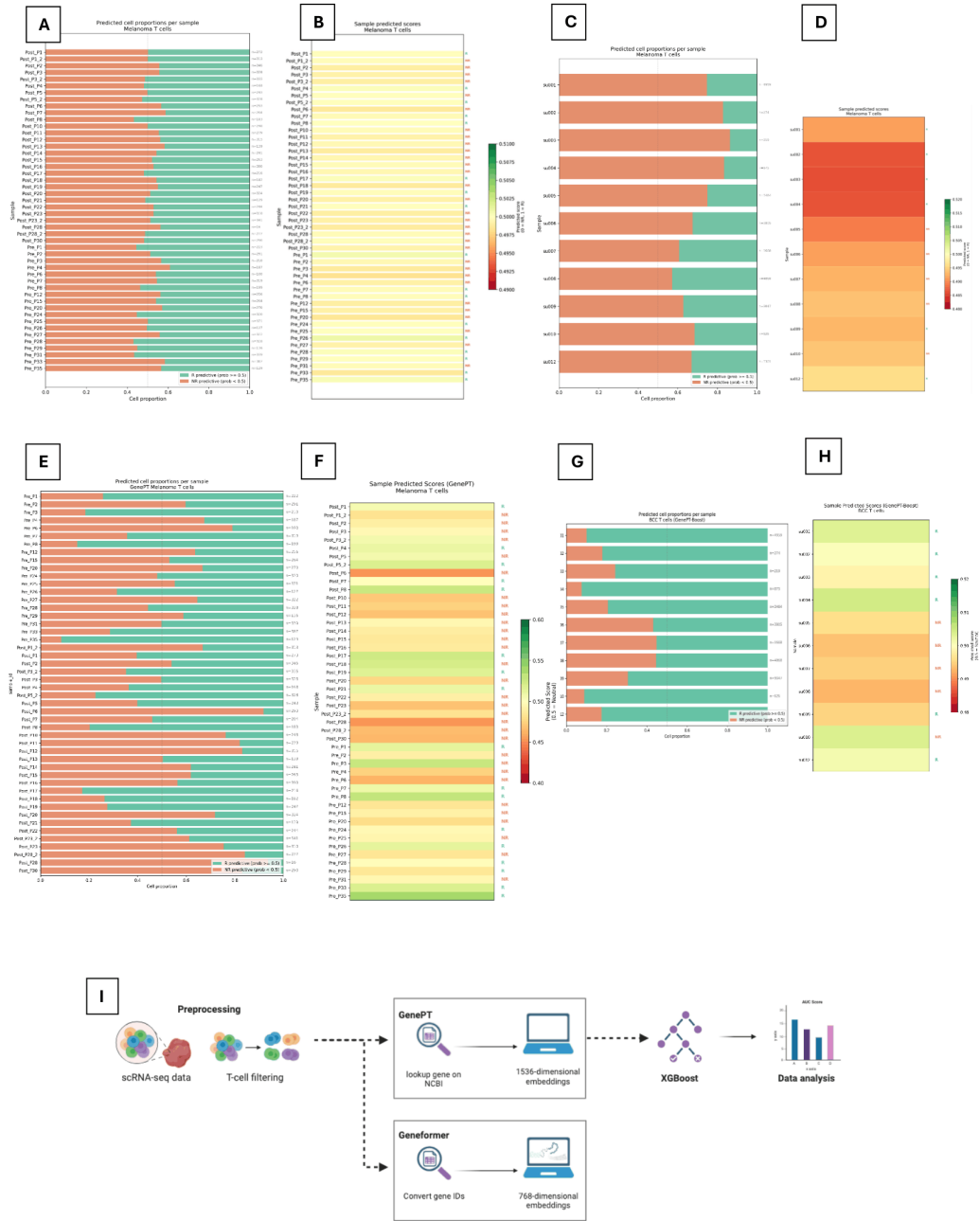


Figure 1: Overview of results from the GG-Boost pipeline. **A.** Geneformer-XGBoost predicted cell proportions per sample from the melanoma cohort. Cells are classified as responders (green, predicted probability ≥ 0.5) or non-responders (red, predicted probability < 0.5). **B.** Sample-level predicted score for the melanoma cohort from Geneformer-Boost predictions. Sample scores were created by using a mean of all cell-level predicted probabilities (methods). Each bar represents one sample coloured on a continuous scale (red = 0.49 to green = 0.51). **C.** Geneformer-Boost predicted cell proportions per sample from the BCC cohort. **D.** Sample-level predicted score for the BCC cohort from Geneformer-Boost predictions. Each bar represents one sample coloured on a continuous scale (red = 0.48 to green = 0.52). **E.** GenePT-Boost predicted cell proportions per sample from the melanoma cohort. **F.** Sample-level predicted score for the melanoma cohort from GenePT-Boost predictions. **G.** GenePT-Boost predicted cell proportions per sample from the BCC cohort. **H.** Sample-level predicted score for the BCC cohort from GenePT-Boost predictions. **I.** Overview of the GG-Boost pipeline.

The BCC dataset consists of 11 samples treated using anti-PD1 ICI therapy and sequenced using droplet-based 10X genomics sequencing technology [14]. After preprocessing (Methods), we were left with 18,675 responder cells and 12,231 non responder cells, originating from 6 responder samples and 5 non-responder samples.

The trained melanoma XGBoost model was applied directly to the BCC Geneformer embeddings, producing per-cell predicted responder probabilities that were aggregated to sample-level scores (Methods). Geneformer-Boost achieved an AUC score of 0.333 (Supplementary Figure S1). Further examination of the distribution of predicted responder probabilities per sample shows that all samples contain mostly non-responder cells (predicted probability < 0.5) (Fig. 1 C). Visualizing the sample-level score, we notice the three samples with the lowest scores are all true responder samples (Fig. 1 D). Additionally, all sample-level scores are less than 0.5 (between 0.486 and 0.499) and thus all classified as non-responders by the models cell-level predictions. This demonstrates that Geneformer-Boost can not identify important features in the BCC embedding space, leading to underconfident predictions. Ultimately Geneformer-Boost is unable to discriminate response status in the BCC context.

The above process was repeated with the BCC GenePT embeddings. GenePT-Boost achieved an AUC score of 0.8 (Supplementary Figure S1). However, the cell specific results were similar to Geneformer-Boost in that the majority of the cells in each sample were classified as responder cells (1 G), while the sample level probabilistic scores ranged between 0.4927 and 0.5046 (1 H), indicating that GenePT-Boost also struggles with uncertain predictions that remain near the decision boundary. Nevertheless, at threshold 0.5 the model achieves sensitivity 0.80 and specificity 0.67, correctly classifying 8 of 11 samples and outperforming PRECISE’s 11-gene signature baseline of 0.68 on this data. However, the AUC is driven by small differences in sample scores, rather than confident separability, reducing the trust we have in its indication of model performance on new data.

4 Discussion

The motivation of our study is to determine whether foundation model embeddings of cells can improve the performance of PRECISE’s classifier on a new cancer type compared to the use of the 11 gene-signature on raw expression data. While GG-Boost achieves high performance (AUC 0.901, 0.905) on the melanoma data set during training and validating, its performance diverges when validating the models on the BCC cancer data set. Geneformer-Boost’s AUC falls to 0.33 and GenePT-Boost’s AUC stays high at 0.80. Both models performed similarly to PRECISE on melanoma data (AUC 0.94), but only GenePT-Boost demonstrates improved performance on new data, achieving an AUC of 0.8 compared to PRECISE’s 11 gene-signature’s AUC of 0.68. Although this seems like a confirmation of our claim, a closer look at sample level scores reveals otherwise.

4.1 GenePT vs Geneformer

First, we examine the differences in performance based on the foundation model used. We argue the difference highlighted above is due to how each model represents gene features, and their differing sensitivity to sequencing platforms.

We can attribute Geneformer-Boost’s failure to generalize primarily to the systematic difference in embedding space due to the difference in sequencing technologies used in either dataset. Geneformer encodes each cell as a rank-ordered sequence of genes, which is sensitive to sequencing depth. Smart-seq2, used in the melanoma dataset achieves a higher read depth per cell, producing a dynamic range of expression values [18]. This allows Geneformer’s gene ranking within each cell based on well-differentiated count values. This is contrasted to 10x Genomics technology used to obtain the BCC dataset which produces a much shallower sequencing depth [19]. This method gives more compressed expression values, where many genes are tied – impacting Geneformer’s rank-value encodings as there is less gene information per cell. Thus, a gene expressed at moderate levels in a Smart-seq2 cell may rank in the top 20% of expressed genes, while the same gene in a 10x cell may rank higher or lower depending on the composition of the detected gene set. This change in rank orderings means that the decision boundaries learned by XGBoost in the melanoma embedding space do not correspond to meaningful biological boundaries in the sparser BCC embedding space, even though both datasets were processed by the same Geneformer model.

GenePT, by contrast, represents each cell as a weighted average of gene vectors derived from NCBI biological texts, where the weights are the expression values of each gene in that cell. Because the weighting is proportional, not rank-based, GenePT embeddings are less sensitive to the absolute number of genes detected: a gene expressed at twice the level contributes twice as much to the embedding regardless of sequencing depth. Moreover, the biological meaning encoded in each gene’s vector is not affected by sequencing platform as it is derived from literature. This is why GenePT embeddings transfer more effectively from Smart-seq2 melanoma data to 10x BCC data than Geneformer embeddings do. Our results therefore provide empirical evidence that literature-based cell representations are more robust to cross-platform transfer than expression-rank-based representations. This suggests future development of foundation models for clinical use may be more generalizable with text based cell representations.

4.2 The Score Concentration Problem

Although GenePT-Boost appeared to generalize better to new cancer, we found that the sample level predictions were concentrated around 0.5. This is due to the weak cell level predictions (also close to 0.5) (Supplementary Figure S2). This narrow spread indicates that the model rarely assigns high-confidence predictions to individual cells. Even the most extreme predictions remain close to the decision boundary of 0.5, suggesting that the classifier is only weakly confident at the cellular level. The high AUC indicates that GenePT-Boost captures the subtle signals in the data, but the differences in probability scores across responders and non responders is very small. Therefore, the high AUC in this setting does not imply meaningful predictive confidence, but a consistent, small trend across cells in a sample. Thus, our model’s clinical usability is limited.

This phenomenon could be due to the idea that many cells within the tumour microenvironment are not directly informative of treatment response [6]. Because all cells inherit the sample label during training, many non-informative cells are considered predictive, adding noise to the model and masking the true signal from predictive cells. Thus, the classifier only learns subtle patterns, yielding weak predicted probabilities near 0.5.

4.3 Comparison with PRECISE

At the surface level, our results confirm the hypothesis that motivated this study. We hypothesised that replacing raw gene expression features with foundation model embeddings would improve cross-cancer generalisation by providing a shared feature space that is more robust to platform differences than raw expression data. On the melanoma training data, both models match PRECISE’s performance, demonstrating that foundation model embeddings preserve sufficient biological signal for immunotherapy response prediction. However, only GenePT-Boost achieved improved cross-cancer generalization over PRECISE’s 11 gene-signature, showing that the full classifier based on literature supported embeddings does in fact outperform an 11 gene heuristic formula, while expression rank based embeddings fail to do the same. These results suggest that foundation model embeddings can improve generalization depending on the model and how sensitive it is to sequencing platforms.

Despite improving XGBoost’s generalization, GG-Boost’s predicted probabilities were weak, suggesting that although we achieved our goal, the quality of the new model’s predictions is subpar. This signals that there is potential for foundation models (especially text-based ones) to improve the generalizability of a model trained on one cancer type to others, but single cell data requires processing to ensure immunotherapy response signals are not diluted.

5 Conclusion

This work is a contribution to the emerging field of foundation model-based clinical prediction models using single-cell data. We present our evaluation of two foundation model embeddings (Geneformer and GenePT) as feature extractors for XGBoost to predict response to ICI treatment. We directly benchmarked against the PRECISE framework at their results on the melanoma and BCC cohorts.

The results suggest that while Geneformer embeddings did not provide a meaningful improvement, GenePT was more promising. By achieving a higher AUC (0.8) on the external validation dataset (BCC), GenePT-XGBoost may be the central focus for future investigation.

Several limitations constrain the conclusions that we can draw. One such is that external validation was only conducted on one cancer type, due to data availability and computational constraints. Validation on the additional datasets used by PRECISE – including TNSBC, NSCLC, and glioblastoma – should be a focus for future research and would substantially strengthen generalizability claims.

Another limitation is that the concentration of sample scores around 0.5 limits the clinical utility of GG-Boost, even if AUC is nominally high, reflecting low prediction confidence at the cell level. One promising approach to address this is the reinforcement learning (RL) cell filtration framework implemented in PRECISE, which identifies and removes non-predictive cells prior to creating foundation model embeddings. The paper demonstrated that this RL-based filtering improved cross-cancer performance in all the data sets, although it only raised the BCC AUC from 0.68 to 0.70. Applying such filtration to the embedding-based pipeline could widen the score distributions and improve both the confidence and generalizability of our pipeline on other cancer data sets.

Another important direction for future work is improving the interpretability of GG-Boost. Interpretability is essential in using ML for medical applications, so future work could explore partial interpretability recovery through back-mapping of important embedding dimensions to contributing genes. For GenePT, embeddings are averages of gene vectors. So, by identifying which dimensions SHAP analysis flags as important for XGBoost predictions and ranking genes by their contribution to those dimensions, it is possible to recover a gene-level importance ranking that can be compared to PRECISE's 11-gene signature.

While language-based foundation models represent a compelling direction for future single-cell representation learning, their application to response prediction is complicated. With thousands of cells per sample, averaging cell-level predictions to a sample score risks diluting the signal from the biologically relevant cell subpopulations that drive response. Future work integrating cell filtration with foundation model embeddings could address these limitations. Ultimately, this work represents an early step towards improving transfer learning for immunotherapy response prediction, establishing a reproducible evaluation framework and highlighting key challenges in cross-cancer generalization that must be addressed before foundation model embeddings can reliably augment or replace raw expression features in clinical prediction pipelines.

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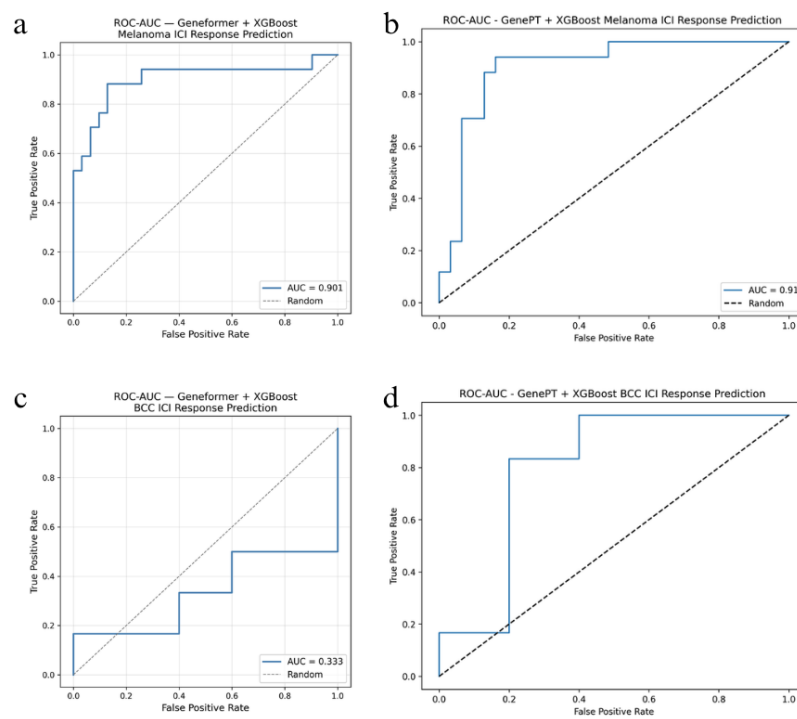
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Appendix

All source code files can be found at https://github.com/yasnaj/Foundation_Model_ICI_Prediction

The contributions of this project and paper were made equally by both Avneet and Yasna. Specifically, all preprocessing of the datasets, pipelines, results, and discussion relating to Geneformer-XGBoost were done by Yasna. And all pipelines, results, and discussion relating to GenePT-XGBoost were made by Avneet.



Supplementary Figure S1: Performance comparison between models

<u>sample_id</u>	min	max	std	mean
Post_P1	0.428437	0.566844	0.041398	0.504750
Post_P10	0.435798	0.540780	0.032281	0.470426
Post_P11	0.434998	0.548590	0.029529	0.474263
Post_P12	0.434372	0.556850	0.030448	0.468224
Post_P13	0.433800	0.564554	0.037685	0.494799
Post_P14	0.433945	0.566564	0.037508	0.488889
Post_P15	0.433429	0.564718	0.038377	0.486632
Post_P16	0.434137	0.559701	0.039497	0.486937
Post_P17	0.435496	0.564392	0.027431	0.523695
Post_P18	0.433047	0.569522	0.034605	0.521138
Post_P19	0.432503	0.567311	0.033015	0.514931
Post_P1_2	0.434939	0.567586	0.036653	0.482219
Post_P2	0.434294	0.562217	0.041081	0.489537
Post_P20	0.433519	0.558955	0.037195	0.477039
Post_P21	0.435210	0.566693	0.038719	0.510415
Post_P22	0.432812	0.560303	0.035501	0.490336
Post_P23	0.433877	0.560564	0.036131	0.468157
Post_P23_2	0.432808	0.559842	0.037842	0.481705
Post_P28	0.437264	0.522263	0.023022	0.453033
Post_P28_2	0.434722	0.553172	0.030875	0.466241
Post_P3	0.432807	0.560149	0.035229	0.496740
Post_P30	0.434807	0.548828	0.031854	0.464710
Post_P3_2	0.433684	0.568502	0.037030	0.506957
Post_P4	0.432893	0.570401	0.032930	0.510282
Post_P5	0.433560	0.570862	0.042425	0.506968
Post_P5_2	0.434223	0.560976	0.027126	0.518348
Post_P6	0.437811	0.548180	0.023879	0.452821
Post_P7	0.432520	0.568039	0.035051	0.500652
Post_P8	0.433274	0.566247	0.032891	0.526438
Pre_P1	0.433837	0.568074	0.031916	0.513584
Pre_P12	0.432582	0.569042	0.036968	0.484019
Pre_P15	0.433667	0.567032	0.039706	0.491129
Pre_P2	0.433556	0.567364	0.036157	0.489858
Pre_P20	0.435214	0.570479	0.039812	0.479604
Pre_P24	0.434240	0.564203	0.039084	0.495507
Pre_P25	0.432839	0.568763	0.036195	0.493605
Pre_P26	0.432998	0.568072	0.035245	0.512813
Pre_P27	0.432442	0.567455	0.038173	0.483775
Pre_P28	0.435857	0.566569	0.032612	0.498609
Pre_P29	0.432928	0.562481	0.040478	0.485567
Pre_P3	0.435748	0.568538	0.033977	0.526968
Pre_P31	0.433637	0.562399	0.040376	0.494872
Pre_P33	0.436013	0.563487	0.031642	0.517977
Pre_P35	0.444283	0.563850	0.023755	0.539625
Pre_P4	0.434148	0.565900	0.039309	0.474524
Pre_P6	0.434968	0.547827	0.034100	0.462950
Pre_P7	0.435164	0.567750	0.032912	0.509139
Pre_P8	0.432587	0.567640	0.035857	0.528175

Supplementary Figure S2: **Ranges of cell specific probabilities for melanoma data - GenePT-Boost.**